

Meridian Petrochemical Industries Ltd.

Equipment Maintenance Report — Centrifugal Pump Unit

Document No.:	MPI-MNT-2026-0417	Report Date:	28 August 2026
Plant / Unit:	Ethylene Cracker Unit - 2	Prepared By:	R. Iyer, Sr. Maintenance Engineer
Facility Location:	Coimbatore Petrochemical Complex	Reviewed By:	S. Menon, Plant Reliability Head
Report Type:	Scheduled Preventive Maintenance	Status:	Completed

1. Equipment Identification

Parameter	Details
Equipment Tag No.	P-204A
Equipment Type	Horizontal Centrifugal Pump
Manufacturer	Sundyne Fluid Systems
Model No.	SDX-4500HC
Service	Naphtha Feed Transfer
Fluid Handled	Naphtha (light hydrocarbon), Flammable Class IB
Design Flow Rate	285 m3/hr
Design Head	142 m
Motor Rating	160 kW, 2980 RPM, 3-phase
Seal Type	API Plan 53B - Dual Mechanical Seal
Installation Date	14 March 2019
Last Overhaul	02 September 2024

2. Inspection Checklist

Item	Inspection Point	Status	Remarks
1	Casing - visual inspection for cracks/corrosion	Pass	No abnormalities observed
2	Mechanical seal - leakage check	Pass	Barrier fluid pressure nominal
3	Bearing housing - vibration analysis	Attention	Vibration 4.8 mm/s RMS, above baseline
4	Bearing lubrication level	Pass	Topped up with ISO VG 32
5	Coupling alignment	Pass	Within 0.05 mm tolerance
6	Suction/discharge pressure gauges	Pass	Calibrated 15-Aug-2026
7	Motor winding insulation resistance	Pass	182 MΩ at 500V

8	Baseplate & foundation bolts	Pass	Torque verified
9	Suction strainer - differential pressure	Attention	ΔP slightly elevated, cleaning scheduled
10	Grounding/earthing continuity	Pass	0.3 ohm, within limit

3. Maintenance History Log

Date	Type	Description	Downtime (hrs)	Technician
14-Mar-2025	Preventive	Routine lubrication and bearing check	1.5	K. Raghavan
02-Jun-2025	Corrective	Mechanical seal replacement due to minor leak	6.0	A. Fernandes
19-Jul-2025	Preventive	Vibration monitoring, alignment check	2.0	K. Raghavan
05-Nov-2025	Corrective	Impeller cleaning, suction strainer replaced	4.5	P. Suresh
21-Jan-2026	Preventive	Quarterly inspection, oil analysis sample sent	1.0	A. Fernandes
28-Aug-2026	Preventive	Scheduled PM - this report	3.0	R. Iyer

4. Findings & Recommendations

During the scheduled preventive maintenance activity on pump P-204A, overall equipment condition was found satisfactory with two items flagged for follow-up action. Bearing housing vibration readings (4.8 mm/s RMS) exceeded the established baseline of 3.5 mm/s RMS, indicating early-stage bearing wear or a minor misalignment condition. It is recommended that a follow-up vibration spectrum analysis be conducted within 30 days to confirm root cause. Additionally, the suction strainer differential pressure was slightly elevated; strainer cleaning has been scheduled for the next planned shutdown window. No leaks, casing defects, or abnormal noise were observed during this inspection. The pump remains fit for continued service under normal operating conditions.

5. Next Scheduled Maintenance

Next PM Due Date	27 November 2026
Follow-up Vibration Check	Within 30 days (by 27-Sep-2026)
Strainer Cleaning	At next unit shutdown
Recommended Spares to Stock	Mechanical seal kit (P/N MS-SDX4500-53B), Bearing set (6316-2RS)

This is a sample/synthetic maintenance report generated for software testing purposes only. All names, tag numbers, and readings are fictitious and do not represent an actual facility.